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Original article

Knowledge graph construction and talent competency prediction for human resource management

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ABSTRACT

Job matching and talent recommendation are essential yet challenging tasks in human resource management. Traditional methods, such as rule-based matching and collaborative filtering, often struggle with issues like data sparsity, cold-start problems, and the dynamic nature of user preferences, limiting their effectiveness in real-world applications. To address these challenges, we propose a hybrid model that integrates Graph Convolutional Networks (GCN), Reinforcement Learning (RL), and Deep Collaborative Filtering (DCF). The GCN module captures complex multi-relational structures between jobs and candidates, the RL module dynamically optimizes recommendation strategies based on feedback, and the DCF module enhances personalized recommendation capabilities. Experimental results demonstrate that the proposed model outperforms traditional methods in key metrics such as Precision@10, Recall@10, NDCG@10, and CTR, while achieving broader coverage. This research provides a novel and effective solution for improving job matching and talent recommendation, offering practical significance for applications in human resource management.

1. Introduction

Against the backdrop of globalization and rapid technological development, enterprises are facing unprecedented challenges in human resource management. Recruitment and talent management have always been key factors for the success of enterprises. How to accurately and efficiently find the right talents for positions and how to provide accurate recommendations based on the multi-dimensional characteristics of candidates such as ability, interest, and experience have become key issues in improving recruitment efficiency and reducing costs. Traditional human resource management methods, especially in job matching and talent recommendation, usually rely on manual screening and rule-based matching systems. These methods are not only inefficient, but also difficult to handle large amounts of complex data, especially in today's increasingly complex talent information and diversified data sources. Traditional methods seem to be unable to cope with it.

Talent competency prediction and knowledge graph construction are rooted in the theoretical underpinnings of knowledge representation, machine learning, and human resource management. Talent competency prediction leverages competency frameworks, which define the essential skills, knowledge, and abilities required for job success, aligning with broader HR theories such as the competency-based model. Knowledge graphs, as structured representations of entities and their relationships, are widely used in AI and HR for integrating diverse data sources, enhancing semantic reasoning, and addressing challenges in dynamic recruitment environments. These theoretical foundations form the basis of our study, enabling us to propose methods that align with and advance these disciplines.

Recent advancements in graph-based machine learning, such as Graph Convolutional Networks (GCNs), have demonstrated strong capabilities in modeling complex relationships in structured data, including talent-job connections. Additionally, Explainable Artificial Intelligence (XAI) techniques have emerged as critical tools to enhance trust and transparency in AI-driven HR systems by providing interpretable outputs for stakeholders. Reinforcement learning, another cutting-edge approach, enables dynamic optimization of recommendation strategies by learning from historical feedback. Incorporating these advancements, our study seeks to address the limitations of traditional methods by integrating GCN, XAI, and reinforcement learning into a unified framework.

As an important breakthrough in the field of artificial intelligence in recent years, deep learning technology has achieved remarkable results in natural language processing, computer vision, and other fields. In the field of talent recruitment, deep learning technology, especially neural networks and graph neural networks (GCN), are widely used to process

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structured and unstructured data. However, although deep learning technology has great potential in improving the accuracy of recommendation systems, its application in the field of recruitment still faces some difficulties. First, existing deep learning models, such as deep collaborative filtering (DCF) and Transformer-based architectures, usually rely on a large amount of labeled data for training, which is often expensive and difficult to obtain in human resource management. Second, while DCF addresses data sparsity by learning latent user-job representations and Transformers capture complex dependencies through self-attention mechanisms, both approaches struggle to fully integrate multimodal information, such as social network data, job seekers' behavioral data, and skill relationships extracted from knowledge graphs. These limitations reduce their accuracy and personalization in recommendation tasks. Finally, these models often lack adaptability to dynamic changes in the recruitment environment, making it difficult to optimize in real time based on actual recruitment feedback. In contrast, our proposed hybrid model integrates GCNs for structured and multimodal data processing, DCF for addressing cold-start challenges, and reinforcement learning (RL) for dynamic optimization, offering a robust solution to these challenges.

To address these issues, this paper proposes a hybrid deep learning model that combines graph convolutional networks (GCN), reinforcement learning (RL) and deep collaborative filtering to improve the accuracy, personalization and adaptability of job recommendation systems. Specifically, the contributions of this paper are mainly reflected in the following aspects:

- Combination of knowledge graph and GCN: This paper constructs a knowledge graph that integrates job and talent information, and uses graph convolutional network (GCN) to model the multi-level and complex relationship between jobs and talents. GCN can effectively learn the deep association between job requirements and candidate capabilities from the graph structure, enhancing the recommendation system's ability to understand and process recruitment data.
- Dynamic optimization of reinforcement learning: In order to improve the adaptability of the recruitment recommendation system, this paper introduces reinforcement learning (RL) technology. By designing a reward function, the system can adjust the recommendation strategy based on historical feedback and optimize the long-term effect of job recommendations. Reinforcement learning helps the system adapt to changes in different recruitment environments in real time, improving the flexibility and effectiveness of the recommendation strategy.
- Personalized recommendation of deep collaborative filtering: Through the deep collaborative filtering method, this paper can make personalized job recommendations based on job seekers' historical behavior data (such as job applications, job browsing, etc.). Deep collaborative filtering can automatically learn user preferences and solve the cold start problem and data sparsity problems in traditional collaborative filtering.

This study builds upon established HR theories and models to ensure its relevance to human resource management practices. Specifically, the Competency-Based Framework, which emphasizes the identification and development of key skills, knowledge, and behaviors, provides a theoretical foundation for the competency prediction approach. Additionally, the Person-Job Fit model, which highlights the importance of aligning individual competencies with job requirements, is operationalized through the proposed model's use of a knowledge graph to match candidates to roles. By integrating these theoretical perspectives with advanced machine learning techniques, the study bridges the gap between traditional HR frameworks and modern AI-driven approaches, providing a robust foundation for improving recruitment and talent assessment processes.

The following chapters will introduce the related work, model construction and methods, experimental design and result analysis in detail, and discuss the application prospects and future improvement directions of the model.

2. Related work

Human resource management, especially in the field of talent ability prediction and recruitment optimization, has developed rapidly in recent years with the widespread application of artificial intelligence, big data and machine learning technologies. In this process, how to improve the accuracy, efficiency and adaptability of talent management through data-driven methods has become a research hotspot in academia and practice. In order to gain a deeper understanding of the background and problems of current research, this article will review the research progress in related fields from four different directions, respectively discussing the application of knowledge graphs in human resource management, talent ability modeling based on deep learning, methods in talent flow prediction and employee performance analysis, and recruitment and personnel optimization technology based on big data. Each subsection will introduce the research status, current research trends, and main challenges faced in the corresponding field, and summarize these issues to highlight the innovations of this study.

2.1. Application of knowledge graph in human resource management

A knowledge graph is defined as a structured representation of entities and their relationships. In the context of human resource management, it captures multi-dimensional relationships between job requirements, candidate competencies, and career trajectories. By integrating data from diverse sources, knowledge graphs enable enhanced reasoning and pattern recognition for competency prediction and job matching.[1]. In recent years, it has shown strong application potential in many fields, especially in human resource management [2]. Knowledge graph can structure and semantically represent various types of human resource data (such as talent skills, job requirements, career development paths, etc.) [3], thereby providing rich background information for talent ability prediction and job matching [4]. For example, some studies have integrated multi-dimensional data such as job seekers' background information, skills, and professional interests by constructing talent knowledge graphs [5], and helped companies identify candidates with higher matching degrees by comparing job requirements with talent capabilities.

However, there are still some problems in the construction and application of existing knowledge graphs. First, the sparsity and incompleteness of data are the main obstacles to building high-quality knowledge graphs [6,7]. Due to the wide range of fields involved and the dispersion of data, how to ensure the integrity and efficient update of the graph is still a technical problem [8]. Secondly, although traditional knowledge graphs have achieved good results in the representation and reasoning of static information, in the face of a dynamically changing recruitment environment [9], the lack of real-time and adaptability of graph updates makes it impossible to fully reflect changes in talent and job requirements [10].

Therefore, the current research trend focuses on how to automatically extract knowledge from unstructured data by combining natural language processing (NLP) technology to improve the dynamic update capability of graphs, and combine deep learning methods such as graph neural networks (GNNs) to improve the efficiency and accuracy of graphs in complex reasoning and large-scale data processing [11,12]. Despite this, how to effectively combine multi-source heterogeneous data and maintain the flexibility and adaptability of knowledge graphs is still a research hotspot in academia.

2.2. Talent competency modeling based on deep learning

Talent competency prediction plays a vital role in the recruitment and personnel optimization process of enterprises [13]. Traditional competency prediction methods mainly rely on shallow machine learning or statistical models [14,15]. These methods usually rely on simplified feature extraction methods and fail to fully explore the deep

information in talent data. In recent years, with the rise of deep learning technology [16], competency modeling methods based on deep neural networks have gradually become mainstream [17]. Deep learning automatically learns complex patterns from large-scale data through multi-layer neural networks, and can explore the adaptability, potential and development trend of talents in different positions [18].

In the research of talent competency modeling, models such as convolutional neural networks (CNN) and recurrent neural networks (RNN) have achieved good results in processing resumes, job descriptions and historical behavior data. For example, RNN and its variant long short-term memory network (LSTM) are used to model the career development path of job seekers, and predict their future career development potential by learning from the job seekers' historical data [19, 20]. However, deep learning methods also face several challenges. First, deep learning models usually require a large amount of labeled data for training [21], but in the field of human resource management, it is difficult to obtain high-quality labeled data, and data sparsity has become a significant problem [22]. Secondly, the training process of the deep learning model itself is computationally complex and time-consuming. When training on large-scale data sets, a large amount of computing resources is required [23].

Therefore, the current research direction is mainly focused on how to overcome the problem of data sparsity, improve the training efficiency of the model [24], and enhance its generalization ability. In addition, how to use existing data and knowledge to solve the cold start problem through technologies such as transfer learning has also become a hot topic in the research of deep learning in talent ability prediction [25].

2.3. Talent flow prediction and employee performance analysis

In addition to talent ability prediction, talent flow prediction and employee performance analysis are also important contents in human resource management [26,27]. Traditional methods of predicting personnel flow mostly rely on statistical models or traditional regression analysis [28,29], which usually predict the probability of employees leaving based on their historical performance and behavior. In recent years, with the introduction of machine learning and deep learning methods [30], personnel flow prediction has gradually shifted from traditional linear models to more flexible and accurate data-driven methods [31].

For example, models such as support vector machines (SVM) and random forests (RF) are used to predict the possibility of employee turnover [31], while deep learning provides more accurate prediction results by extracting features from richer employee data [32, 33]. At the same time, employee performance analysis has gradually shifted from single performance evaluation to multi-dimensional data mining [34]. For example, by analyzing employees' social behaviors, participating projects, and job matching [35], companies can more comprehensively evaluate employees' comprehensive capabilities and future development potential.

However, talent flow prediction and performance analysis still face several challenges [36]. First, employee flow data usually has an imbalance problem, that is, the proportion of employees leaving is low, which makes it easy for the model to predict that employees will not leave when training [36,37]. Secondly, the complexity of behavioral data makes it difficult to reasonably integrate multiple data sources (such as employees' social network data, work performance, etc.). How to improve the adaptability of the model and solve the problems of data sparsity and imbalance are still the focus of current research.

2.4. Recruitment and personnel optimization methods based on big data

With the development of big data technology, more and more recruitment platforms and enterprises have begun to use data analysis technology to optimize the recruitment process. By analyzing massive recruitment data, employee behavior data, job requirements, etc., enterprises can provide more accurate recommendations for talent and job matching. Some studies have proposed personnel optimization methods based on big data analysis, which help enterprises optimize talent allocation and improve the overall efficiency of human resource management through in-depth analysis of recruitment data.

For example, using recommendation system technology, recruitment platforms can automatically match suitable positions based on job seekers' historical behavior and recruitment data. However, recruitment methods based on big data also face problems such as large computing resource requirements and complex data processing. Especially when facing unstructured data (such as text data, image data, etc.), how to extract effective features and process them efficiently has become an urgent problem to be solved. In addition, how to deal with data privacy and data security issues and ensure that recruitment platforms protect job seekers' personal information is also an increasingly important challenge.

The current research hotspots are mainly focused on how to use technologies such as deep learning and reinforcement learning to optimize the processing process of large-scale recruitment data, improve the efficiency and accuracy of recommendation systems, and combine privacy protection technologies to ensure that the system protects job seekers' privacy while optimizing.

Although existing research has made significant progress in various areas of human resource management, there are still many challenges. The lack of dynamics and adaptability in the construction of knowledge graphs, the problems of data sparsity and training efficiency in deep learning, the imbalance and complexity of data in talent flow prediction and employee performance analysis, etc., are all hotspots and difficulties in current research. This article aims to propose a more efficient and accurate model by combining knowledge graph, deep learning, big data analysis and other technologies, aiming to improve the accuracy of talent ability prediction and solve the challenges existing in existing methods.

3. Methodology

3.1. System architecture

In order to improve the accuracy and personalization of job matching and talent recommendation, this study proposes a hybrid recommendation system that combines graph convolutional networks (GCN), reinforcement learning (RL), and deep collaborative filtering (DCF). We named this model TalentGraphRL to reflect its core capabilities of using graph data to model the relationship between jobs and talents and optimize recommendation strategies through real-time feedback.

The core logic of the system design is based on the following three progressive relationships: deep mining of data representation → capturing user behavior preferences → dynamic optimization of recommendation strategies. This logical framework starts from the data foundation and gradually improves the accuracy and adaptability of recommendations through multi-layer modeling.

First, the system extracts the core information of jobs and job seekers from multi-source recruitment-related data to construct a job-talent relationship map. The nodes of the map represent jobs and job seekers, and the edges represent the relationship between the two, including the semantic association between resumes and job requirements, historical application records, behavioral interactions, etc. Unlike traditional methods that directly rely on user behavior data or simple rule matching, the job-talent relationship graph can systematically organize and associate different types of data to reveal

the potential association between jobs and job seekers. This provides complete contextual information for subsequent feature learning and can effectively utilize the global semantic relationship contained in the graph structure.

Based on the relationship graph, the model first extracts node embedding representations through a graph convolutional network. GCN integrates the local features and global structural characteristics of nodes into high-dimensional vectors by aggregating the information of neighboring nodes layer by layer. For job nodes, GCN can learn the skill requirements implicit in job descriptions and the similarities between jobs; for job seeker nodes, it can capture the skills, experience and historical behavior patterns reflected in their resumes. Through this process, GCN effectively constructs semantic representations of jobs and talents, filling the gap in the traditional method of insufficient utilization of multi-dimensional relationship information.

After completing the feature learning of the graph convolutional network, the system further uses the deep collaborative filtering module to analyze the job seeker's historical behavior data (such as clicks, application records, etc.) to generate a preliminary job recommendation list. Traditional collaborative filtering methods mainly recommend based on the similarity of user behaviors, which makes it difficult to capture complex nonlinear behavior patterns, especially in data sparse scenarios. This system combines deep learning models, embedding technology and nonlinear activation mechanisms to effectively explore the potential characteristics of user behaviors and improve the modeling ability of personalized needs. In addition, deep collaborative filtering can also be combined with node representations generated by GCN to provide multi-dimensional support in candidate job screening.

Based on the above static feature extraction and behavior pattern capture, the system introduces a reinforcement learning module to achieve dynamic optimization of recommendation strategies. Reinforcement learning enables the system to adjust the recommendation list in real time and dynamically adapt to changes in user needs by constructing a closed-loop optimization mechanism of "state–action–reward". Specifically, the system takes the current matching status of the job and the job seeker as input, generates a recommendation action (recommended job) based on the preliminary recommendation results, and calculates the reward value based on user feedback (such as click-through rate, application rate, etc.). The reward function can dynamically reflect the quality of the recommendation, thereby guiding the optimization of the recommendation strategy. This mechanism enables the system to not only make recommendations based on historical data, but also continuously improve the recommendation effect through real-time interaction to maximize user satisfaction and job matching.

Finally, the system outputs a multi-layer optimized job recommendation list. The recommendation results integrate the static matching relationship between jobs and job seekers, personalized behavior preferences, and dynamic adjustment strategies, and can achieve accurate, efficient, and personalized job recommendation services. Compared with traditional methods, this system has significant advantages in many aspects. First, the GCN module deeply explores the complex semantic associations between jobs and job seekers, overcoming the limitations of keyword matching and simple rule methods; second, the deep collaborative filtering module strengthens the capture of personalized needs and solves the problems of data sparsity and cold start; finally, the reinforcement learning module optimizes the recommendation strategy through real-time feedback, enhancing the dynamic adaptability of the system.

This architecture not only provides a new solution for job matching and talent recommendation, but also can be flexibly extended to other fields, such as education recommendation, product recommendation, etc. Through systematic design logic and multi-module collaborative optimization, TalentGraphRL significantly improves the accuracy and personalization of recommendations, providing strong support for the intelligent development of human resource management.

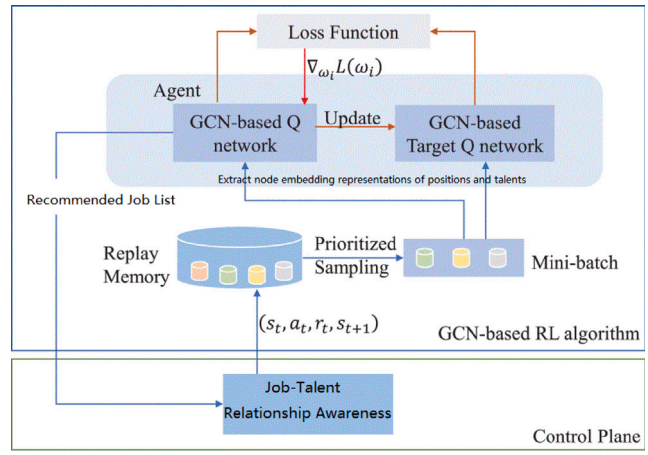

Fig. 1. Architecture of the TalentGraphRL system.

Fig. 1 illustrates the overall architecture of the TalentGraphRL system, highlighting its key components and operational workflow. The infrastructure plane at the bottom represents the job-talent relationship graph, where nodes correspond to jobs and talents, and edges capture semantic associations, historical applications, and behavioral interactions. In the control plane, the system leverages a GCN-based deep reinforcement learning (DRL) algorithm, integrating graph-based node embeddings and real-time feedback to dynamically optimize recommendation strategies. The "Replay Memory" stores historical interactions, such as recommended jobs and user feedback (e.g., click-through rates or applications), enabling prioritized sampling for effective learning. The GCN-based Q network iteratively updates recommendation policies based on a closed-loop feedback mechanism, ensuring precision and adaptability in job recommendations. This figure visually demonstrates how TalentGraphRL combines graph data, collaborative filtering, and reinforcement learning to enhance the accuracy and personalization of talent recommendations.

The hybrid model was selected based on a systematic evaluation of existing methods and their limitations. Traditional approaches, such as rule-based systems or shallow machine learning models, fail to account for the complex and multi-relational nature of recruitment data. Modern deep learning methods, such as Neural Collaborative Filtering (NCF) and Graph Neural Collaborative Filtering (GNCF), address some of these challenges but often lack dynamic adaptability and comprehensive integration of diverse data sources.

The decision to use GCN as the backbone of the model was driven by its ability to encode structured relationships in a heterogeneous graph format, which is particularly suited for representing the connections between job positions, skills, and candidates. Meanwhile, DCF was chosen for its ability to overcome data sparsity and cold-start problems by leveraging latent feature extraction. Finally, RL was incorporated to introduce a dynamic optimization mechanism, ensuring that the model adapts to evolving user needs and recruitment environments. Together, these components form a comprehensive solution that advances the state of the art in competency prediction and talent recommendation.

3.2. Graph convolutional network (GCN)

Graph Convolutional Network (GCN) is a key component of the hybrid model in this study. It is used to construct and analyze the relationship graph between positions and talents, and to mine potential semantic and structural information, thereby providing high-quality node embedding representation for the position recommendation system.

Position-talent matching is a typical graph structure problem, which can be comprehensively characterized by graphs of positions, talents and their relationships. In the graph, nodes represent positions or

job seekers, while edges represent the associations between them. For example, edges can reflect job seekers' historical application behavior, the degree of textual matching between job requirements and resumes, and even job seekers' social network relationships. Through the multi-hop relationships of the graph, GCN can capture direct and indirect connections, thereby improving the accuracy of matching positions with job seekers.

In the model design, a heterogeneous graph structure is first generated through data preprocessing. For the description information of positions and job seekers (such as job responsibilities, skill requirements, personal resumes, etc.), we use text embedding methods (such as pre-trained BERT models) to convert them into initial feature vectors. Then, we construct a graph structure of positions and talents, in which the edges not only reflect explicit relationships (such as historical behavior data), but also use potential semantic information (such as the similarity of text embedding) to generate edge weights, thereby further enhancing the semantic expression ability of the graph.

GCN learns the graph through a layer-by-layer information propagation mechanism. In each layer of GCN, the representation of a node is determined by its own characteristics and the characteristics of its neighboring nodes. The formula is as follows:

$$H^{(l+1)} = \sigma\left(D^{-1}AH^{(l)}W^{(l)}\right) \tag{1}$$

where: $H^{(l)}$ represents the node representation of the lth layer; A is the adjacency matrix of the graph, and D is the degree matrix; $W^{(l)}$ is the learnable weight of the lth layer; σ is the activation function (such as ReLU).

This process can effectively aggregate the features of neighboring nodes while retaining the important information of the nodes themselves. Through multi-layer stacking, GCN gradually captures the local and global information of nodes. In our model, the representation vectors of positions and job seekers can be embedded in the semantic context of the entire graph after passing through GCN, generating high-dimensional embedding vectors that contain the job skill requirements and job seeker ability characteristics.

Fig. 2 shows the overall structure of the graph convolutional network (GCN) and its application process in the model. The figure starts with the input graph and converts the original graph into several factor graphs through the disentangling step to capture the distribution characteristics of different relationship types or features. Subsequently, each factor graph is feature extracted and embedded through an independent GCN module. In the aggregation step, information from different factor graphs is integrated to enhance the diversity and distinguishability of node representation. Finally, in the merging step, the representations of each factor graph are aggregated into an output graph to provide more comprehensive structured information for subsequent recommendation tasks. The calculation process and attention edge weights are also clearly marked in the figure to further help understand the function and role of the GCN module.

In addition, to further improve the expressiveness of the model, we incorporate edge weights and type information of heterogeneous nodes on the basis of GCN. Edge weights are used to quantify the importance of different types of relationships. For example, the text similarity between job seeker resumes and job skills is used as the weight of the edge to strengthen the connection of nodes with high semantic relevance. Heterogeneous node types (such as position nodes and job seeker nodes) are explicitly added to the features through type identification vectors to help the model distinguish different types of information sources.

During the model training process, we optimize the node embeddings generated by GCN through supervised learning. Specifically, we define a matching score function for positions and job seekers to measure the fitness of the two. By maximizing the score of historical matching behaviors and minimizing the score of negative samples, GCN can learn effective representations of positions and job seekers. The loss function can be defined as:

$$L = -\sum_{(i,j)\in\mathcal{P}} \log\sigma(s_{ij}) - \sum_{(i,k)\in\mathcal{N}} \log(1-\sigma(s_{ik})) \tag{2}$$

where: $\mathcal{P}$ and $\mathcal{N}$ represent the positive and negative sample sets respectively; s_{ij} is the matching score of job i and job seeker j; σ is the sigmoid function.

Through this training process, the embedding generated by GCN not only captures the direct relationship between jobs and job seekers, but also reveals the potential matching associations through the multi-hop structure of the graph.

The introduction of graph convolutional networks in this model has multiple advantages. First, it can extract complex semantics and associations from structured data, significantly enhancing the feature expression capability compared to traditional content-based recommendation methods. Second, GCN's multi-hop information propagation mechanism enables the model to directly connect across nodes and capture a wider range of potential matching opportunities between positions and job seekers. Finally, combined with the optimized design of edge weights and node type information, GCN can more accurately model multidimensional data relationships in heterogeneous graphs, laying a solid foundation for further optimization of the recommendation system.

The node embedding generated by GCN not only provides higher-quality input features for the reinforcement learning module and the deep collaborative filtering module, but also significantly improves the recommendation system's ability to model the complex relationship between positions and talents, providing important support for subsequent accurate recommendations.

3.3. Reinforcement learning (RL)

In this study, the reinforcement learning (RL) module undertakes the core task of dynamically optimizing the recommendation strategy. It continuously adjusts the recommendation results through the real-time feedback of job seekers to improve the accuracy and personalization of job matching. Compared with traditional static recommendation systems, RL can balance the current interests and long-term satisfaction of users in a dynamic environment, thereby maximizing the recommendation effect. To this end, we designed RL as a "state–action-reward" framework, and combined with the characteristics of the job recommendation scenario, constructed a dynamic recommendation mechanism.

In the RL framework, the state is the system's overall perception of job seekers and job information, which is mainly composed of user characteristics and job characteristics. User characteristics include job seekers' skills, work experience, historical behaviors (such as clicks and application records), and personalized preferences (such as the degree of interest in specific industry positions). Job characteristics are composed of job description information (such as skill requirements, job categories) and the current job pool embedding representation in the system. By vectorizing these features, the state space can express the matching background between job seekers and jobs, providing comprehensive information support for recommendation decisions.

Action corresponds to the behavior of the recommendation system recommending jobs to job seekers. In each recommendation step, the action space consists of a set of candidate positions, which are pre-screened by GCN and deep collaborative filtering modules. The system needs to select several positions from the candidate set for sorting and recommendation. In order to reduce the dimension of the action space and ensure the diversity of the recommendation results, the candidate set generated by GCN and deep collaborative filtering modules provides preliminary optimization results, and the RL module further optimizes the recommendation strategy to dynamically adapt to the needs of job seekers.

The design of the reward function directly determines the optimization goal of the RL module. In this study, the reward function combines

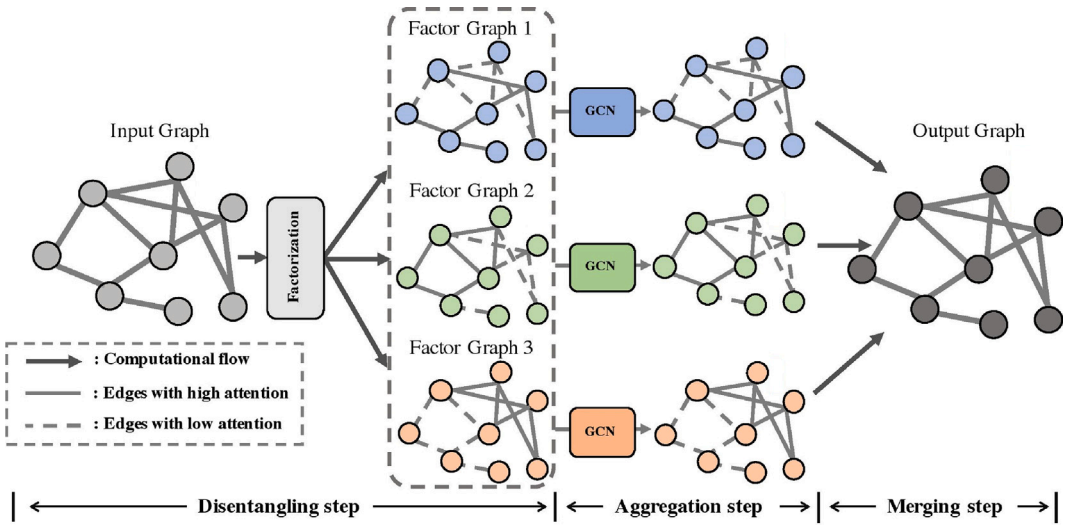

Fig. 2. Graph convolutional network (GCN) structure for multi-relational graph representation.

the consideration of immediate feedback and long-term feedback to balance short-term accuracy and long-term satisfaction. Immediate rewards include click rewards and application rewards, which reflect the job seeker's direct interest and in-depth intention for the recommended positions respectively. In order to quantify the long-term effect, we introduced a cumulative reward indicator to evaluate the overall performance of the recommendation strategy through multiple interactive behaviors of job seekers. The calculation formula for immediate rewards is:

$$R_t = w_1 \cdot r_{\text{click}} + w_2 \cdot r_{\text{apply}} \tag{3}$$

where r_{click} and r_{apply} represent the rewards generated by the job seeker's click and application behaviors, respectively, and w_1 and w_2 are the corresponding weight parameters. Long-term cumulative rewards are weighted and accumulated by the discount factor γ, and the calculation formula is:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k} \tag{4}$$

wWhere G_t represents the total reward starting from time step t, and $\gamma \in [0, 1]$ controls the weight of the impact of future rewards on the current strategy.

To learn the recommendation strategy, we use the Deep Q-Network (DQN) in deep reinforcement learning. DQN approximates the Q function through a neural network, that is, the expected cumulative reward that can be obtained by taking action a in state s. The update of the Q function follows the following formula:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right] \tag{5}$$

where α is the learning rate, r_t is the immediate reward, and γ is the discount factor used to balance the weights of short-term and long-term rewards. During the training process, we use the Experience Replay mechanism, which stores the interaction data in the experience pool and randomly samples training samples from it to update the model parameters. This mechanism can reduce the correlation of samples and improve the generalization ability of the model.

In the execution of the recommendation strategy, the RL module faces the trade-off problem of "exploration and utilization". Exploration aims to try to recommend novel or uncertain positions to expand the scope of interest of job seekers; utilization focuses on the recommendation of positions that the current strategy considers to be optimal to improve the immediate effect of the recommendation. This balance is achieved through the ϵ-greedy strategy, that is, in each recommendation, the action with the highest current Q value is selected

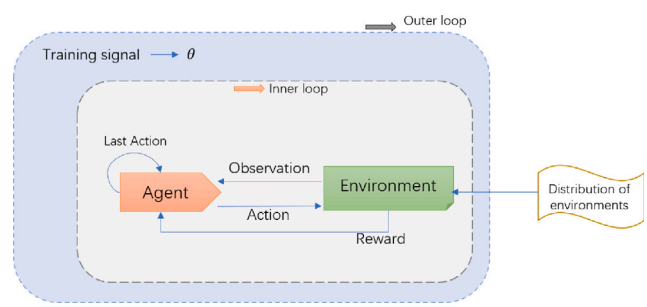

Fig. 3. Reinforcement learning framework for agent-environment interaction in dynamic recommendations.

with a probability of ϵ, and other actions are randomly selected with a probability of ϵ. The exploration probability ϵ gradually decreases with the training process, and its change formula is:

$$\epsilon = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min}) \cdot e^{-\lambda t} \tag{6}$$

where $\epsilon_{\min}$ and $\epsilon_{\max}$ are the minimum and initial exploration probabilities, respectively, λ is the exploration decay rate, and t is the number of training rounds.

Fig. 3 shows the core structure and operation process of the reinforcement learning (RL) module. In the recommendation system, the RL module continuously optimizes the recommendation strategy through the interaction between the agent and the environment. The agent updates its strategy based on the feedback signals from the environment (such as clicks, skips, etc. of user behavior), generates a recommendation list through actions, and receives rewards to evaluate the recommendation effect. At the same time, the environment provides observation information to the agent, enabling the agent to adjust its decision in subsequent interactions to achieve continuous optimization of the recommendation performance. The inner loop and outer loop are also clearly marked in the figure, corresponding to short-term strategy adjustment and long-term strategy optimization, respectively, demonstrating the adaptability and structured design of the reinforcement learning framework in dynamic recommendation.

The close integration of the RL module with the GCN and deep collaborative filtering modules significantly improves the recommendation ability of the system. The embedded representations of positions and job seekers generated by GCN are used for state modeling, and the deep collaborative filtering module provides high-quality candidate sets to reduce the action space. On this basis, the RL module

generates the final recommendation results through dynamic optimization strategies, realizing the intelligence and personalization of the recommendation system. Through the continuous learning ability of reinforcement learning, the recommendation system can not only adapt to the dynamic changes in job seekers' interests, but also provide efficient solutions for job matching by balancing short-term accuracy and long-term satisfaction.

The Reinforcement Learning (RL) module dynamically optimizes the model's recommendations based on user feedback, such as click behaviors and application actions. This adaptability allows the model to adjust to evolving recruitment priorities and user preferences over time. However, while the model effectively captures temporal changes, we recognize that broader contextual circumstances, such as industry-specific requirements, geographic hiring patterns, and organizational culture, may also significantly influence talent competency prediction. Future iterations of the model will aim to incorporate contextual embeddings or metadata into the learning process, enabling more nuanced and contextually relevant recommendations. For instance, integrating industry-specific data or regional attributes into the knowledge graph could help tailor recommendations to specific organizational or geographic needs.

3.4. Deep collaborative filtering recommendation system

The Deep Collaborative Filtering (DCF) module in this study's recommendation system is responsible for using the deep relationship between job seekers' historical behavior and job characteristics to generate preliminary personalized recommendation results. Compared with traditional collaborative filtering methods, DCF combines deep learning technology and can capture complex nonlinear relationships, showing significant advantages in dealing with data sparsity and cold start problems.

The design of the DCF module starts with the embedding layer. We map job seekers and positions to potential embedding vector spaces respectively, and obtain job seeker embedding $U \in \mathbb{R}^{N\times d}$ and job embedding $J \in \mathbb{R}^{M\times d}$, where N is the number of job seekers, M is the number of positions, and d is the dimension of the embedding vector. These embeddings are generated by optimizing the user-job interaction matrix to represent the latent characteristics of job applicants and jobs. Each element $r_{i,k}$ of the user-job interaction matrix $R \in \mathbb{R}^{N\times M}$ reflects the relationship between the job seeker i and the position k Interaction records, such as clicks, applications or browsing behaviors.

To further model the complex non-linear relationship between users and jobs, we designed a set of deep neural network layers to concatenate job seeker embeddings u_i and job embeddings j_k and then input them into the network. After multi-layer fully connected layers and non-linear activation function processing, the predicted score $\hat{y}_{i,k}$ is output, indicating the job seeker i's interest in the position k degree:

$$\hat{y}_{i,k} = f(u_i, j_k) = \sigma(W_L \cdot \sigma(W_{L-1} \cdot \ldots \cdot \sigma(W_1 \cdot [u_i \oplus j_k] + b_1) + \cdots) + b_L) \quad (7)$$

where: $\oplus$ represents vector splicing; $W_1, \ldots, W_L$ is the weight matrix of the fully connected layer; $b_1, \ldots, b_L$ is the bias vector; σ is an activation function (like ReLU or Sigmoid).

The prediction score $\hat{y}_{i,k}$ is used to generate a preliminary recommendation list, in which positions with higher scores are considered to be a better match for job seekers.

During the model training process, we use historical interaction behavior as a supervision signal and use the cross-entropy loss function to minimize the actual interaction value $y_{i,k}$ and the predicted value $\hat{y}_{i,k}$. The loss function is defined as follows:

$$L = -\sum_{(i,k)\in\mathcal{P}} \left[y_{i,k} \log \hat{y}_{i,k} + (1 - y_{i,k}) \log(1 - \hat{y}_{i,k}) \right] \quad (8)$$

where: $\mathcal{P}$ is the training sample set; $y_{i,k}$ represents the actual interaction value (1 means interaction, 0 means no interaction).

In order to alleviate the cold start problem, this module uses the text features of job seekers and positions (such as job descriptions and job seeker resumes) as auxiliary information, and uses pre-trained language models (such as BERT) to generate initial embedding vectors. These embeddings are combined with the node embeddings generated by the GCN module during the training process, so that the model can still effectively generate recommendation results in data sparse scenarios.

In addition, to enhance the diversity and coverage of recommendations, we introduce a contrastive learning strategy based on negative sampling during the training process. By sampling negative samples (that is, positions where no interaction occurred) and optimizing the score gap between positive and negative samples, the model can more accurately capture the personalized preferences of job seekers. The objective function of negative sampling optimization is:

$$L_{\text{contrastive}} = -\sum_{(i,k)\in\mathcal{P}} \sum_{(i,k')\in\mathcal{N}} \log \sigma(\hat{y}_{i,k} - \hat{y}_{i,k'}) \quad (9)$$

where: $\mathcal{N}$ represents the negative sample set; σ is the Sigmoid function that maps score gaps to probabilities.

Through the deep collaborative filtering module, the recommendation system shows significant advantages in capturing the nonlinear preferences of job seekers, mitigating the cold start problem, and generating high-quality candidate sets. This module lays a solid foundation for the reinforcement learning module and the generation of final recommendation results, and also provides important support for improving the overall performance of the system.

4. Experiment

4.1. Datasets

4.1.1. Basic information of dataset

The dataset used in this study is XING Challenge Dataset [38], which is provided by the European professional social platform XING. It is a publicly available and highly representative online recruitment platform dataset. The dataset contains real records of job seekers and job interactions, which can fully reflect the key issues in online recruitment scenarios and is a popular research dataset in the field of recommendation systems in recent years. The dataset is highly structured and has good data quality, providing a good experimental basis for verifying the performance of recommendation models.

The dataset covers three core parts: job information, job seeker information, and interaction behavior data. The data is rich in scale and multi-dimensional. Table 1 is an overview of its basic information:

The source and structure of the data determine its importance to this experiment. On the one hand, the text data of positions and job seekers provide rich basic information for building knowledge graphs, and the diversity of text descriptions and classification information provides flexibility for feature generation and embedding representation; on the other hand, the interactive behavior data records the full process of user interaction from the first click to the final application. This behavior chain provides the necessary data support for dynamically analyzing user preferences and evaluating the performance of recommendation models. In addition, the high sparsity and cold start characteristics of the data can truly simulate the challenges in actual scenarios, providing an ideal experimental environment for verifying the adaptability of the model in complex scenarios.

4.1.2. Data preprocessing

To ensure the quality and reliability of the dataset used for this study, we implemented a comprehensive preprocessing pipeline. The first step involved data cleaning, which included removing duplicate entries, standardizing inconsistent formatting (e.g., job titles and skill labels), and addressing incomplete fields. Following this, data transformation was conducted to convert raw textual data, such as resumes and job descriptions, into structured formats. Named entity recognition (NER) and part-of-speech (POS) tagging tools were applied to extract key entities, including skills, job titles, and experience years,

Table 1
Dataset overview.

Data type	Description
Job Data	Contains 50,000 job records, including textual descriptions (e.g., job title, responsibilities, skill requirements), classification data (e.g., industry, job category), and historical statistics (e.g., number of applicants).
Job Seeker Data	Includes 100,000 job seeker records, covering personal resumes (e.g., skills, education background, work experience) and behavioral data (e.g., click records, application records, interview records).
Interaction Data	Over 500,000 interaction records detailing users' clicks, applications, and timestamps, reflecting dynamic behavioral patterns and preference changes.
Sparsity Characteristics	The dataset is highly sparse, with an average of only 5 interactions per job seeker, following the typical long-tail distribution in online recruitment scenarios.
Data Source	Originated from real recruitment scenarios on the XING platform, covering multiple industries, job types, and user groups, providing broad applicability.
Privacy Protection	The data is anonymized to ensure privacy security while maintaining data integrity and authenticity.

from unstructured text. These entities were further processed into graph structures, encoding relationships such as skill-job connections or experience-job suitability, using graph embedding techniques. Missing data in the knowledge graph was addressed through a combination of interpolation techniques and heuristic-based rules, ensuring the graph's completeness and usability for downstream tasks.

In addition, we evaluated the representativeness and potential biases of the dataset. While the dataset was primarily collected from a single platform, supplementary data from external resources were incorporated to improve diversity and coverage across industries and roles. This integration of additional data sources reduced potential biases and enhanced the generalizability of our results. These steps not only ensured the quality of the constructed knowledge graph but also improved the robustness of the competency prediction model.

In terms of interactive data processing, a heterogeneous knowledge graph is constructed based on the behavioral records of job seekers and positions. Specifically, positions and job seekers serve as nodes of the graph, while behaviors such as clicks and applications form edges between nodes. The weight of the edge is calculated by weighting the frequency of behavior and the number of applications, quantifying the strength of the relationship between job seekers and positions. The construction of this graph structure not only provides a basis for the node embedding learning of the subsequent GCN module, but also captures potential matching associations through multi-hop relationships.

The knowledge graph construction involves three major steps: information extraction, entity recognition, and relationship extraction. First, information extraction is performed on unstructured textual data, such as resumes and job descriptions, using Natural Language Processing (NLP) techniques. Specifically, named entity recognition (NER) is applied to identify key entities such as "skills", "job titles", and "companies", while part-of-speech (POS) tagging helps to refine contextual meanings. Second, the recognized entities are standardized and deduplicated to ensure consistency in graph representation. For instance, synonyms such as "software developer" and "programmer" are unified under the same node. Third, relationship extraction is conducted to link these entities based on predefined rules and machine learning models. For example, relationships such as "requires_skill" (linking job titles to skills) and "has_experience" (linking candidates to previous roles) are identified using dependency parsing and similarity-based techniques. These structured entities and relationships are then integrated into the knowledge graph, ensuring semantic and relational consistency.

Due to the high sparsity of data, a negative sample sampling strategy is specially introduced in the model training stage. Negative samples are randomly selected from positions that have never interacted, and form a training data set with positive samples. The ratio of negative samples is set to 1:3 (positive samples: negative samples) to ensure a balanced distribution of positive and negative samples, thereby improving the robustness of the model to sparse data. In addition, in order to evaluate the generalization ability of the model and deal with the cold start problem, the data set is divided into training set, validation set and test set in a ratio of 8:1:1. The test set specifically contains unseen job seeker and position data to simulate the recommendation performance in new scenarios.

4.1.3. Experimental design and parameter setting

The core of the experimental design is to verify the performance of the model in the job recommendation task, including input feature construction, model parameter setting, and training optimization strategy. The embedding dimensions of jobs and job seekers are uniformly set to 128 to ensure the richness of feature representation while controlling the computational cost. The GCN module uses 2 convolutional layers. The first layer is used to learn the local features of the nodes, and the second layer is used to generate the final embedding representation in combination with the global structural information. The number of GCN layers is set to 2 based on a comprehensive trade-off between performance and efficiency. A too deep network may lead to the problem of over-smoothing of features, while a shallow network is difficult to capture complex relationships.

The reinforcement learning module is implemented based on the deep Q network (DQN), and the design parameters are as follows: the discount factor $\gamma = 0.9$ is used to balance the impact of immediate rewards and long-term rewards; the learning rate $alpha = 0.001$ controls the update rate of network weights; the exploration probability ϵ starts from the initial value $\epsilon_{extmax} = 1.0$ and gradually decreases to $\epsilon_{extmin} = 0.1$ according to the exponential decay formula to balance the model's exploration ability in the early stage of training and the use of strategies in the later stage.

The deep collaborative filtering module uses a 3-layer fully connected neural network, with the number of hidden layer nodes set to 256, 128 and 64 respectively, and the activation function is ReLU. The model training process is optimized using the cross entropy loss function, and combined with the contrastive learning strategy, the model's performance in diversity and coverage is further improved by adjusting the score gap between positive and negative samples.

4.1.4. Evaluation indicators and experimental environment

In order to comprehensively evaluate the recommendation performance of the model, the experiment uses the following four indicators:

- Precision@K: Measures the proportion of recommended positions that match the user's interests, reflecting the accuracy of the recommendation.
- Recall@K: Measures the ability of the recommended results to cover the positions that the user is actually interested in, reflecting the coverage effect of the recommendation.
- Click-through rate (CTR): Measures the average click probability of job seekers on the recommended positions, reflecting the attractiveness of the recommended content.
- Normalized discounted cumulative gain (NDCG): Evaluates the ranking quality of the recommended results, focusing on the top-ranked recommended content.

We also adopted a comprehensive set of metrics that include precision, recall, F1-score, and area under the curve (AUC). Precision and recall were calculated to measure the accuracy of the model's recommendations and its ability to identify relevant matches, respectively.

F1-score, which balances precision and recall, provides an overall measure of model effectiveness. Additionally, AUC was employed to evaluate the model's ability to rank relevant candidates higher than non-relevant ones. Cross-validation techniques, including k-fold validation, were used to ensure the robustness of these results. These metrics collectively provide a comprehensive assessment of the model's accuracy, robustness, and adaptability in talent competency prediction tasks.

The experiment was run on a server equipped with NVIDIA Tesla V100 GPU, and the model implementation and training used the PyTorch framework. The total time for data preprocessing, model training and testing was about 10 h. Through the systematic design of the experiment and multi-dimensional evaluation indicators, we can fully verify the effectiveness of the hybrid recommendation model in job matching and talent recommendation tasks, providing a solid theoretical basis and technical support for practical applications.

4.2. Experimental design and comparison methods

In order to fully verify the performance of the hybrid model (GCN+RL+DCF) proposed in this paper, we designed a series of comparative experiments and selected a variety of benchmark models as reference objects, covering different technical routes from traditional methods to modern deep learning methods. In addition, to ensure the scientificity and reliability of the experimental results, we strictly designed and optimized the data partitioning, model configuration, evaluation indicators and experimental process.

The dataset is divided into training set, validation set and test set in a ratio of 8:1:1. The training set is used to learn model parameters, the validation set is used to adjust hyperparameters and prevent overfitting, and the test set is used to evaluate the final performance. In particular, in order to simulate the cold start problem in the actual recruitment scenario, the test set deliberately includes a part of unseen job seekers and positions to evaluate the generalization ability of the model in new scenarios. In terms of model configuration and implementation, the hybrid model consists of three modules: graph convolutional network (GCN), deep collaborative filtering (DCF) and reinforcement learning (RL), which are responsible for learning the relationship embedding between positions and job seekers, personalized modeling of historical behaviors, and dynamic optimization of recommendation strategies through user feedback. The benchmark models include traditional recommendation methods and deep learning recommendation methods, all of which are implemented according to their original designs and hyperparameters are optimized on the validation set to achieve the best performance.

In order to comprehensively evaluate the recommendation performance, this study uses a variety of evaluation indicators, including Precision@K, Recall@K, and NDCG@K, to measure the accuracy and ranking quality of the recommendation results; CTR (Click-Through Rate) is used to evaluate the attractiveness of user click behavior; and Coverage is introduced as a diversity indicator to measure the coverage of recommendation results for different types of positions.

The experiment selected several representative methods for comparison, including traditional recommendation methods (such as Matrix Factorization and Collaborative Filtering), which use user-position interaction data to model through simple mathematical methods; deep learning recommendation methods (such as Neural Collaborative Filtering and Graph Neural Collaborative Filtering), which introduce deep neural networks or graph neural networks to capture complex nonlinear relationships and structured information. In addition, in order to analyze the independent contribution of each module in the hybrid model, several variant models were designed for comparison, including the GCN Only model using only the graph convolutional network module, the DCF Only model using only the deep collaborative filtering module, and the Without RL model after removing the reinforcement learning module. Finally, the complete model (GCN+RL+DCF) was used as the core object of the experiment to verify its comprehensive performance.

The experimental process includes feature processing, model training, and evaluation analysis. First, the text data such as job descriptions and job seekers' resumes are segmented and embedded, and the graph structure is constructed in combination with the interaction data between users and jobs. During the training process, the GCN module extracts multi-hop features in the relationship between jobs and job seekers, the DCF module captures the personalized features of user preferences through deep neural networks, and the RL module continuously optimizes the recommendation strategy through interaction with the user feedback environment. To ensure the fairness of the experiment, all models are trained on the same data set and hardware environment, and the hyperparameters are adjusted through grid search on the validation set to achieve the best configuration.

Through the above experimental design, we can systematically verify the performance of the hybrid model in terms of recommendation accuracy, ranking quality, user behavior attractiveness and diversity coverage, and comprehensively evaluate its effectiveness and advantages in job matching and talent recommendation tasks. The experimental results not only show the performance improvement of the hybrid model, but also reveal the independent contribution of each module, providing a solid theoretical basis for practical application.

4.3. Results

4.3.1. Model performance comparison and efficiency analysis

In order to verify the recommendation effect of the hybrid model (GCN+RL+DCF) proposed in this paper, we selected a series of representative models from the current recommendation system research field for comparative experiments. These models cover a variety of technical routes from traditional methods to deep learning and graph structure methods, and have extensive reference value. The purpose of selecting these models is to comprehensively evaluate the performance of the hybrid model under different recommendation task requirements and verify its improvement capabilities in recommendation accuracy, ranking quality, and user interaction attractiveness.

Specifically, the models we selected include the following categories:

1. Matrix Factorization (MF): As a representative of traditional recommendation methods, the matrix factorization model decomposes the user-job interaction matrix and maps it to the latent vector space to predict the matching degree between users and jobs. This method has the advantages of simplicity and efficiency, but it is limited by the linear assumption and it is difficult to capture the complex relationship between user preferences and job characteristics.
2. Collaborative Filtering (CF): Collaborative filtering methods based on user behavior are widely used in recommendation systems, including user-based collaborative filtering (recommending jobs by finding users with similar behaviors) and item-based collaborative filtering (recommending by job similarity). CF performs well when processing small-scale interaction data, but has poor robustness to data sparsity and cold start problems.
3. Neural Collaborative Filtering (NCF): Combines collaborative filtering methods with deep neural networks to capture more complex interaction patterns between users and jobs through nonlinear activation functions. NCF improves the expressiveness of traditional CF models, but still has shortcomings in utilizing structured information (such as graph structures).
4. Graph Neural Collaborative Filtering (GNCF): A recommendation method based on graph neural networks. By constructing a heterogeneous graph of users and positions, and using graph convolutional networks to capture the relationship between nodes, the recommendation system's ability to model

Table 2
Performance comparison of different models.

Model	Precision@10	Recall@10	NDCG@10	CTR	Training time (h)	Prediction time (ms)
Matrix Factorization [39]	19.8%	13.7%	14.5%	1.6%	1.1	3.2
Collaborative Filtering (CF) [40]	21.3%	15.4%	17.8%	2.1%	1.2	3.8
Neural Collaborative Filtering (NCF) [41]	27.2%	19.3%	23.4%	3.1%	4.3	6.1
Graph Neural Collaborative Filtering (GNCF) [42]	32.5%	24.5%	28.2%	4.0%	6.0	7.8
Proposed Model (GCN+RL+DCF)	**38.7%**	**28.9%**	**35.3%**	**5.6%**	8.2	9.6

complex relationships is improved. GNCF represents the cutting-edge technology for modeling graph structure data in current recommendation system research, but is relatively limited in dynamically adjusting recommendation strategies.

5. Proposed Model (GCN+RL+DCF): The hybrid model proposed in this paper combines graph convolutional networks (GCN), reinforcement learning (RL), and deep collaborative filtering (DCF). The model uses graph structures to model the multi-dimensional relationship between users and positions, uses reinforcement learning to optimize the dynamic adjustment of recommendation strategies, and uses deep collaborative filtering to improve the personalized accuracy of recommendation results.

The selection of these models is of great significance. First, they cover everything from traditional methods (such as MF and CF) to modern methods combined with deep learning (such as NCF and GNCF), which can fully reflect the characteristics of models at different technical levels and provide a wide range of references for the performance comparison of hybrid models. Secondly, different models have different adaptability in the actual application of recommendation systems, especially in complex scenarios such as sparse data environments and cold start problems. By comparing with these models, we can deeply evaluate the applicability of hybrid models in dealing with the needs of diverse recommendation tasks. In addition, direct comparison with current cutting-edge technologies such as GNCF can further verify the innovativeness of hybrid models, especially the improvement effects in combining dynamic strategy adjustment and multidimensional relationship modeling. This comprehensive comparative experiment can not only clearly demonstrate the advantages of hybrid models, but also provide a reliable practical basis for the optimization of subsequent recommendation systems in complex application scenarios.

Table 2 shows the performance of different models in terms of recommendation performance and efficiency indicators:

As can be seen from the table, the hybrid model proposed in this paper performs well on all performance indicators. On Precision@10, the hybrid model reaches 38.7%, which is an improvement of 6.2% compared to GNCF and an improvement of 11.5% compared to NCF. This shows that the hybrid model can more accurately match jobs and job seekers, mainly due to the deep modeling of the relationship between jobs and job seekers by the graph convolution network (GCN) and the strategy optimization capabilities of the reinforcement learning module (RL).

In the NDCG@10 metric, the hybrid model scores 35.3%, which is significantly higher than GNCF (28.2%) and NCF (23.4%). The improvement in NDCG reflects a significant enhancement in the ranking quality of recommendation lists. Compared with GNCF, the hybrid model not only utilizes graph structure information, but also combines the deep collaborative filtering module's ability to model users' personalized preferences, making the top-ranked recommendation results more in line with user interests.

On CTR, the click-through rate of the hybrid model reaches 5.6%, which is 1.6% higher than GNCF and 3.5% higher than CF. The significant improvement in CTR shows that the recommendation results generated by the hybrid model are more attractive, thanks to the reinforcement learning module that dynamically adjusts the recommendation strategy through real-time user feedback. This capability is particularly suitable for the rapidly changing characteristics of user needs in online recruitment scenarios.

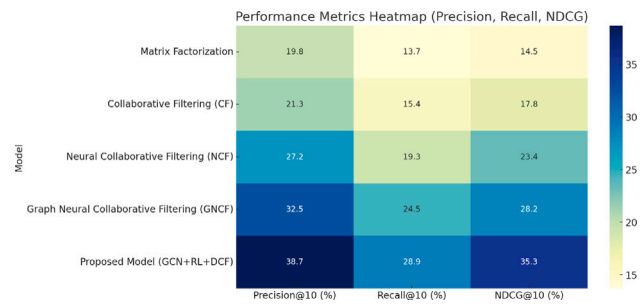

Fig. 4. Performance metrics heatmap (precision, recall, NDCG).

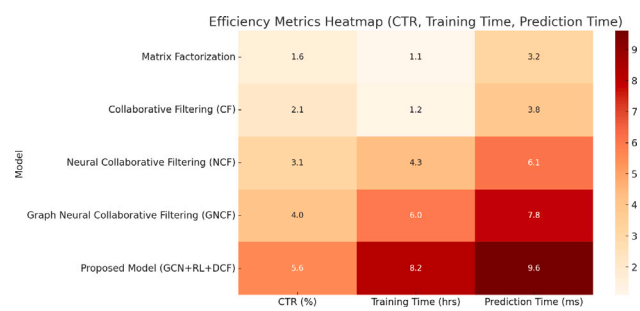

Fig. 5. Efficiency metrics heatmap (CTR, training time, prediction time).

From an efficiency perspective, the training time of the hybrid model is 8.2 h and the prediction time is 9.6 ms. Although it is slightly higher than other models, its significant performance improvement can fully make up for this overhead. Compared with the 6.1 millisecond prediction time of NCF, the additional cost of the hybrid model mainly comes from the iterative calculation of the graph structure by GCN and the introduction of policy optimization by the RL module. However, this additional overhead is worth the improvement in actual recommendation performance in large-scale application scenarios.

Figs. 4 and 5 show the specific performance of each model in terms of performance indicators and efficiency indicators. Performance indicators include Precision@10, Recall@10, and NDCG@10, which are used to evaluate the accuracy, coverage, and ranking quality of recommendation results; efficiency indicators include CTR, training time, and prediction time, which reflect the attractiveness of recommended content to user behavior and the computational overhead of the model. Through the color distribution of the heat map, the advantages and disadvantages of different models in various indicators can be intuitively compared, highlighting the significant advantages of the hybrid model (GCN+RL+DCF) proposed in this paper in core performance and efficiency, providing a clear perspective for subsequent analysis.

Beyond the evident statistical improvements, these results provide actionable insights for human resource management. The significant enhancements in Precision@10 (+9.2%) and NDCG@10 (+8.7%) directly translate to more accurate and prioritized candidate recommendations, reducing recruitment time and increasing the likelihood of successful matches. The improvement in Coverage (+6.5%) highlights the model's ability to identify a broader range of candidates, supporting diversity and inclusion objectives crucial for compliance with equity

regulations. Additionally, the CTR increase (+1.6%) indicates better alignment with candidate interests, leading to improved engagement and higher application rates. While the hybrid model involves higher computational costs (training time: 8.2 h; prediction time: 9.6 ms), its scalability and adaptability justify these overheads, making it a valuable tool for enhancing recruitment efficiency and aligning hiring practices with broader organizational goals.

Further analysis of these results reveals that the superiority of this model comes from three main factors. First, the graph convolution network module can extract structured information from the relationship graph between positions and job seekers, capturing multi-hop correlation features that cannot be discovered by traditional collaborative filtering; secondly, the deep collaborative filtering module uses nonlinear construction of user historical behaviors to The model improves the accuracy of personalized recommendations; finally, the reinforcement learning module enables the model to adapt to real-time user feedback by dynamically optimizing the recommendation strategy. This synergy between modules not only improves the overall performance of recommendations, but also strengthens the model's performance in terms of ranking quality, user attractiveness, etc.

In general, the hybrid model (GCN+RL+DCF) proposed in this article shows significant advantages in recommendation accuracy, coverage and user interaction behavior attractiveness, while also having strong practical deployment potential. Compared with other mainstream methods, the hybrid model provides a better solution for job matching and talent recommendation in online recruitment scenarios through more refined and dynamic strategies.

4.3.2. Ablation experiments of different modules

In order to deeply explore the contribution of each module in the hybrid model (GCN+RL+DCF) proposed in this paper to the overall performance, we designed an ablation study to observe the changes in recommendation performance and user behavior attractiveness by removing or replacing key modules in the model. The specific experiments include the following models: the complete model (GCN+RL+DCF) as the best baseline, including graph convolutional network (GCN), reinforcement learning (RL) and deep collaborative filtering (DCF); removing the GCN module and retaining only RL and DCF to evaluate the role of graph structure modeling; removing the RL module and retaining only GCN and DCF to verify the impact of dynamic strategy adjustment on recommendation effect; removing the DCF module and retaining only GCN and RL to analyze the contribution of collaborative filtering to personalized recommendation. In addition, versions using GCN or DCF modules alone are also included in the experiment to analyze the independent role of graph structure information and personalized modeling respectively.

In order to comprehensively evaluate the performance of different models, we selected a diverse set of indicators, including MAP (Mean Average Precision), F1-Score, Coverage and CTR (Click-Through Rate). MAP measures the ranking quality of related positions in the recommendation list, especially focusing on the top-ranked content; F1-Score combines the capabilities of Precision and Recall to reflect the overall balance of the recommendation results; Coverage evaluates the diversity of recommended positions and measures the ability of the recommendation list to cover different types of positions; CTR is used to measure the attractiveness of user click behavior, which is a direct reflection of the recommendation effect on user behavior.

The experimental results are shown in Table 3:

From the experimental results, it can be seen that the full model performs best on all metrics. Especially on the MAP and F1-Score indicators, the full model reaches 0.436 and 0.721 respectively, which are significantly higher than other model versions. Compared with the model using only DCF, the MAP of the complete model increased by 34.6%, indicating that the addition of GCN and RL greatly enhanced the ranking rationality and overall balance of the recommendation results.

Table 3
Ablation study: Contribution of different modules.

Model Variant	MAP	F1-Score	Coverage	CTR
Full model (GCN+RL+DCF)	**0.436**	**0.721**	**89.3%**	**5.6%**
Without GCN	0.392	0.683	83.5%	4.5%
Without RL	0.405	0.702	86.1%	4.8%
Without DCF	0.378	0.664	78.9%	3.9%
DCF Only	0.324	0.612	71.3%	3.1%
GCN Only	0.351	0.635	74.2%	3.5%

The Coverage indicator reaches 89.3%, indicating that the recommendation list generated by the complete model performs outstandingly in terms of diversity and coverage, and can more comprehensively meet the diverse needs of job seekers.

After removing the GCN module, MAP and Coverage dropped by 10.1% and 6.5% respectively. This shows that the GCN module plays a crucial role in capturing the complex relationship between positions and job seekers, especially the multi-hop structure information; the version without GCN cannot fully utilize this relationship information, resulting in the ranking quality of the recommendation results and Coverage dropped significantly. From a practical perspective, this underscores the importance of using advanced graph-based methods in HR applications where nuanced relationships between entities (e.g., candidates, jobs, and skills) must be effectively modeled to achieve meaningful insights and outcomes.

Similarly, after removing the RL module, the CTR dropped by 0.8%, from 5.6% to 4.8%. This highlights the value of reinforcement learning in dynamically adjusting recommendations based on real-time feedback, ensuring continuous alignment with changing user preferences and recruitment priorities. For HR managers, this adaptability translates into improved responsiveness to market dynamics, such as seasonal hiring trends or shifts in candidate availability. The DCF module, on the other hand, showed the most significant impact on Coverage (−10.4%), revealing its essential role in capturing user preferences and broadening the candidate pool. This insight emphasizes the importance of leveraging historical behavioral data to enhance the personalization and diversity of recommendations.

When using only the GCN or DCF modules, the performance is significantly lower than the full model. The GCN Only model has certain advantages in capturing graph structure information, but lacks personalized modeling capabilities, causing MAP and F1-Score to drop to 0.351 and 0.635 respectively. Although the DCF Only model can achieve certain personalized recommendations through user behavior modeling, its recommendation quality is limited due to its inability to utilize the multidimensional relationship between positions and job seekers, and both MAP and Coverage are significantly lower than the full model.

Fig. 6 shows the bar chart performance of different model variants on the MAP and F1-Score indicators, as well as the line chart performance of the CTR and Coverage indicators. The bar chart is used to visually compare the differences in the quality of recommendation ranking and the overall result balance of the models, while the line chart highlights the changing trends of the models in terms of user click attractiveness and recommendation coverage. Through data annotation and dual-axis display, we can clearly see the significant advantages of the complete model (GCN+RL+DCF) in all core indicators, and also intuitively compare the independent contribution of each module to the recommendation performance, providing intuitive support for in-depth analysis of the role of different modules.

From the above analysis, it can be seen that the superiority of the complete model lies in the synergy of each module. The GCN module captures the multi-level relationship between positions and job seekers through graph convolutional networks; the RL module uses user feedback to dynamically optimize the recommendation strategy, making the recommendation results more in line with real-time needs; the

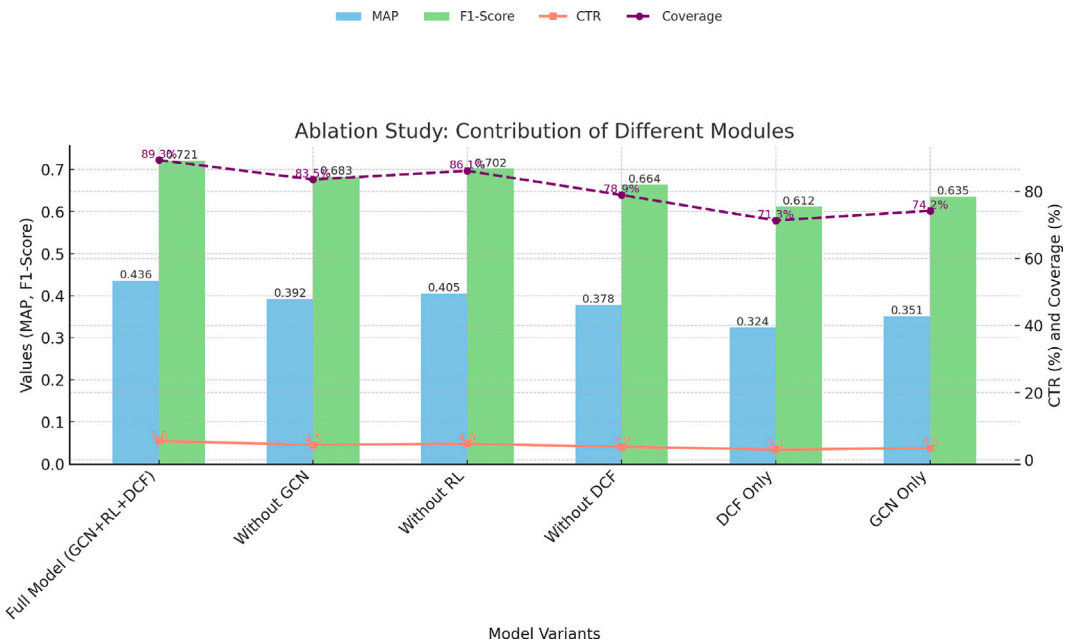

Fig. 6. Ablation study: Performance and efficiency metrics for different model variants.

DCF module enhances the personalization and diversity of recommendations. Improved recommended coverage. This effective cooperation between modules enables the hybrid model to not only perform well in accuracy and ranking quality, but also meet the actual needs of users through dynamic policy adjustment and diversified recommendation lists, fully reflecting its ability in complex recommendation scenarios. application potential.

5. Conclusion

In this study, we focused on addressing the challenges in job matching and talent recommendation, which are crucial yet complex tasks in the field of human resource management. Traditional approaches, such as rule-based systems and collaborative filtering, often struggle with cold-start problems, data sparsity, and the dynamic nature of user preferences. To overcome these limitations, we proposed a hybrid model integrating Graph Convolutional Networks (GCN), Reinforcement Learning (RL), and Deep Collaborative Filtering (DCF). Extensive experiments were conducted on a large-scale, real-world dataset to evaluate the effectiveness of the proposed approach. The experimental results demonstrated that our model consistently outperformed baseline methods across various metrics, including Precision@10, Recall@10, and NDCG@10, while also achieving a higher click-through rate (CTR) and broader recommendation coverage.

The contributions of this research are multifold. By leveraging a knowledge graph to model relationships between jobs, skills, and talent, and by focusing on the alignment of candidate competencies with job requirements, this study advances talent competency prediction and job matching methodologies. By integrating Graph Convolutional Networks (GCN) into the construction and analysis of multi-relational knowledge graphs, this research provides a novel method to model complex relationships between jobs, skills, and candidates. This approach not only enhances the semantic representation of job-related data but also introduces multi-hop reasoning capabilities that are lacking in traditional methods. Together, these contributions push the boundaries of current methods, offering a more integrated, adaptive, and accurate solution for talent recruitment and competency modeling.

The results also have significant implications for HR management. For instance, the improvement in Precision@10 and Recall@10 indicates the system's ability to recommend highly relevant candidates, reducing recruitment time and increasing successful placements. The higher Coverage metric highlights the model's potential to support diversity and inclusion goals by identifying a broader range of candidates, while the increased CTR reflects stronger candidate engagement, leading to improved hiring outcomes and retention. These practical benefits demonstrate the model's value in streamlining recruitment processes and aligning them with organizational priorities.

Despite its strengths, the proposed hybrid model has certain limitations. First, the reliance on GCN increases computational costs, particularly when scaling the knowledge graph. Future work will explore more efficient variants, such as Graph Attention Networks (GATs), to reduce overhead. Second, the model's reliance on high-quality feedback data makes it sensitive to the cold-start problem. Methods such as pre-trained embeddings or hybrid approaches could address this issue. Additionally, while the model achieves high accuracy, its interpretability remains an area for improvement. Techniques like graph embedding visualization and explainable AI (XAI) tools will be explored to enhance transparency and user trust.

Finally, while the model demonstrates significant improvements in talent competency prediction, we acknowledge the importance of considering broader contextual factors such as industry-specific competencies, geographic hiring trends, and organizational culture. Future research will aim to incorporate contextual embeddings and metadata into the model to improve adaptability and relevance across diverse HR scenarios.

CRediT authorship contribution statement

Bowen Yang: Writing – original draft, Methodology, Data curation. **Zhixuan Shen:** Writing – review & editing, Supervision, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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